

YEAR 7 · UNITS 2 AND 3 · HOMEWORK 8

Where It Lands on the Line

Science 2.8 and 2.7 – the pH scale; compounds and mixtures

Mathematics 2.6, 2.5 and 3.1–3.2 – inequalities, equations, rounding

Section F – the pH scale, the cylinder and the graph are one thing

Name: _____

Class: _____

Date given: _____

Date due: **Monday 28 September 2026****How to do this homework**

Six sections. Science first this time: the maths in Section C reads the science table in Section A.

- In science, write **full English sentences**. “Acid” is not a sentence; “Lemon juice is an acid because its pH is below 7” is.
- In maths, **write the equation before you write the answer**. **No calculator**, except where Section E says so.

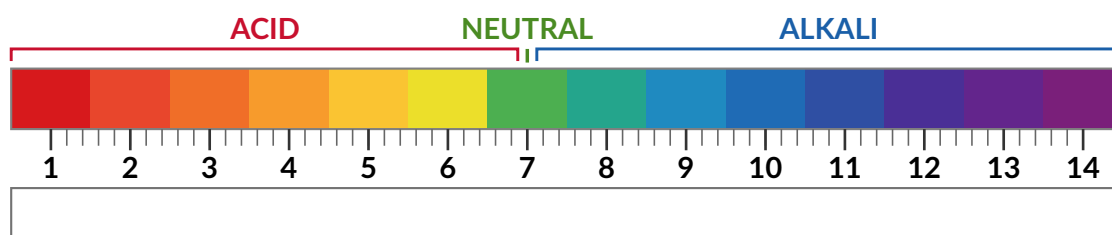
Section A · Science 2.8 – Acids, Bases and the pH Scale

A1 Put each liquid on the scale

Below is the pH scale, coloured the way universal indicator colours it.

Word help – five words the whole section runs on

acid	a substance that tastes sour. A strong one burns skin
base · alkali	the chemical opposite of an acid. An alkali is a base dissolved in water , and it is the word the exam uses
neutral	not an acid and not an alkali. pH exactly 7
indicator	a substance that changes colour to show an acid or an alkali



(a) Every pH below is a **decimal**. **Draw a line** on the scale where it goes and **write the letter**

in the white strip – there are **five small spaces** between each pair of numbers. **(b)** Finish the last column.

	Liquid	pH	Acid, neutral or alkali?		Liquid	pH	Acid, neutral or alkali?
A	lemon juice	2.4		E	sea water	8.2	
B	orange juice	3.8		F	baking soda	9.6	
C	black coffee	5.2		G	soap	10.8	
D	pure water	7.0		H	bleach	12.6	

A2 Why the school buys both

Litmus is cheap. Universal indicator costs more. Mr Bowen buys both.

Two colours, or a number

Litmus has **two** colours only: red in an acid, blue in an alkali. **Universal indicator** gives a **different colour for every pH number**, so it gives you a number instead of just a word.

Mr Bowen tests two liquids with **red litmus paper**. Both turn it blue. One is pH 8 and the other is pH 13. Explain why litmus **cannot** tell him which is which, and say which indicator he should use instead.

Start: "**Litmus can only show...**" then "**...because both liquids are...**"

A3 The end that nobody worries about

A student says: "*Acids are the dangerous ones. pH 13 is safe, because 13 is a big number.*"

Mr Bowen keeps a bottle of oven cleaner under his kitchen sink. Its pH is **13**. Explain why the student is wrong. Use the word **corrosive**.

A4 Neutralisation

What neutralisation does

Add a **base** to an **acid** and they cancel each other out. The acid **stops being an acid** – the pH moves **towards 7**.

(a) Mr Bowen's garden soil has a pH of 4, and rice will not grow in it. He must move the pH **up** to about 7. Tick the **one** thing he should dig into the soil.

What he could add	Tick one	What he could add	Tick one
vinegar, pH 3		lime powder, pH 12	
lemon juice, pH 2		pure water, pH 7	

(b) Explain your choice in full sentences. Say what happens to the pH of the soil.

Section B · Science 2.7 – Compounds and Mixtures

B1 Compound or mixture?

Tick one box in each row.

The test is whether you can get it back

A **mixture** is stirred together, so you can **separate** it again – with a magnet, a filter, or by boiling. A **compound** is **bonded** together: it is a **new substance**, and a magnet will not undo it.

	Compound	Mixture		Compound	Mixture
air			water, H ₂ O		
salt, NaCl			sea water		
sand stirred into salt			carbon dioxide, CO ₂		
mineral water			iron sulfide, FeS		

B2 Iron and sulfur, twice

Mr Bowen puts iron filings and yellow sulfur powder in a dish and **stirs** them. Then he puts a second lot in a test tube and **heats** it until it glows.

He holds a magnet over each one. The magnet pulls the iron out of the **first** dish. It does nothing at all to the **second**.

Explain, in full sentences, what heating did and why the magnet stopped working. Start: "**Be-fore it was heated, the iron and the sulfur were...**" then "**Heating made the atoms...**"

Section C · Mathematics 2.6 — Inequalities

C1 Say it in English

Write $<$ or $>$ in each box, then write the whole thing as an English sentence. The first one is done for you.

Word help – the words a question uses instead of the signs

is less than $\cdot <$ also fewer than, below, under. *Under 18* means age < 18

is greater than $\cdot >$ also more than, above, over. **Greater** is not **great**

The numbers	$<$ or $>$	Say it in English
6 11	$<$	6 is less than 11
9 2		
-4 1		
-7 -3		

C2 Which liquids does the inequality mean?

Here are the eight liquids from A1 again, with their pH.

A	B	C	D	E	F	G	H
lemon juice	orange juice	black coffee	pure water	sea water	baking soda	soap	bleach
2.4	3.8	5.2	7.0	8.2	9.6	10.8	12.6

The letter p stands for the pH of a liquid. For each inequality, write the letters of **every** liquid that works.

The number in the inequality never works itself

$p > 9.6$ means **greater than** 9.6. A liquid at **exactly** 9.6 does **not** count. This is the open circle: the circle is empty because that number is left out.

One of the six has **no answer at all**. If you find it, write **none**.

(a) $p < 7$

(b) $p > 7$ _____

(c) $p > 9.6$ _____

(d) $p < 3.8$ _____

(e) $p > 5$ and $p < 9$ _____

(f) $p > 12.6$ _____

C3 Now write the inequality

Each sentence is a real rule. Write it using $<$ or $>$. Use the letter given.

(a) A swimming pool is safe when its pH is **above** 7.2. Use p . _____

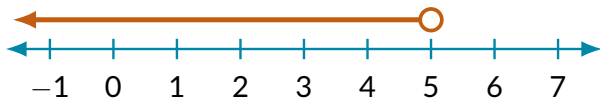
(b) Rain is called acid rain when its pH is **below** 5.6. Use r . _____

(c) Rice grows when the soil pH is **more than** 5. Use s . _____

(d) A lab liquid is corrosive when its pH is **under 2 or over 12**. Use c . Write **two** inequalities. _____

C4 Read the line, then draw the line

(a) Write the inequality each number line shows. Use x .

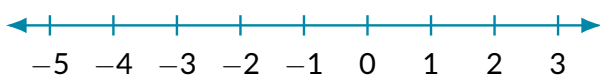


(i) _____



(ii) _____

(b) Now draw this one. Open circle, then an arrow.



$$m > -2$$

C5 The smallest one that works

An **integer** is a whole number: negative, zero or positive.

(a) $x > 6$. Smallest integer? _____

(c) $n > -1$. Smallest integer? _____

(b) $y < -4$. Largest integer? _____

(d) $t > 2.5$. Smallest integer? _____

Section D · Mathematics 2.5 — Equations and Formulae

D1 Put it back in and see

Someone has already written an answer. Put it back into the equation. Tick **right** or **wrong**, and if it is wrong, write the correct answer.

Checking is one line of work

To check $x - 7 = 5$ when somebody says $x = 2$, **put 2 in:** $2 - 7 = -5$, not 5. So it is wrong.

Equation	Their answer	Right	Wrong	If wrong, write it
$x - 7 = 5$	$x = 2$			
$3y = 24$	$y = 8$			
$m + 9 = 15$	$m = 24$			
$k - 6 = 10$	$k = 4$			

D2 Two steps

Undo the **last** step first. Write the middle line.

Last on, first off

$2a + 4 = 18$. The last thing done to a was **add 4**, so undo that first.

$18 - 4 = 14$, then $14 \div 2 = 7$. Check: $2 \times 7 + 4 = 18$. **Not $18 \div 2$ first.**

- | | | | |
|--------------------|-------------|-------------------|-------------|
| (a) $3a + 5 = 26$ | $a =$ _____ | (d) $2d - 9 = 15$ | $d =$ _____ |
| (b) $4b - 7 = 25$ | $b =$ _____ | (e) $6m + 4 = 40$ | $m =$ _____ |
| (c) $5c + 12 = 47$ | $c =$ _____ | (f) $8n - 3 = 45$ | $n =$ _____ |

D3 You are given the formula. Find the letter.

In each one you are told the **answer** the formula gives. Work backwards to find the letter. **Write the equation first.**

Two jobs, in this order

A taxi costs $C = 15n + 20$ where n is the number of kilometres. Mr Bowen's taxi cost **95**. How far did he go?

1. Put the number in: $15n + 20 = 95$

2. Solve it: $95 - 20 = 75$, then $75 \div 15 = 5$. He went 5 km.

(a) $P = 4s$ (perimeter of a square, side s). $P = 52$. $s =$ _____

(b) $C = 8n + 15$ (cost of n tickets). $C = 95$. $n =$ _____

(c) $d = 60t$ (km travelled in t hours). $d = 480$. $t =$ _____

(d) $V = 30h$ (cm^3 in a tank at depth h cm). $V = 420$. $h =$ _____

(e) $m = 60 + v$ (grams: a beaker plus $v \text{ cm}^3$ of water). $m = 145$. $v =$ _____

(f) $F = 2n - 6$. $F = 38$. $n =$ _____

D4 Word problems

Write the equation, then solve it. Show your working in the box.

(a) Mr Bowen's jug holds 340 cm^3 of water. He pours the **same amount** into each of **6** beakers, and 40 cm^3 is left in the jug. How much water went into each beaker?**(b)** Mr Bowen has some fish. He buys **4** more. Then his cat eats **half** of all of them. He now has **6** fish. How many did he have at the start?**Section E • Mathematics 3.1–3.2 – Place Value and Rounding****E1 Move the digits**

No calculator.

The point never moves – the digits do 7.2×10^3 is **7200**. **Not 7.2000**.Multiplying moves the digits **left**. Dividing moves them **right**. The power counts **how many places**.

(a) $3.7 \times 100 =$ _____ (c) $26 \times 10^4 =$ _____ (e) $63 \div 10^3 =$ _____

(b) $4.5 \times 10^3 =$ _____ (d) $940 \div 100 =$ _____ (f) $5 \div 10^4 =$ _____

E2 Round it**Keep the zero**

34.9892 to **1 decimal place** is **35.0**, not 35. The .0 is what shows you rounded to one place. Writing 35 loses the mark.

To 1 decimal place	7.34	12.87	0.952	29.96
To 2 decimal places	5.376	0.0961	8.2449	3.997

E3 One place further**Work out one extra digit, then round**

For $58 \div 7$ correct to **3 decimal places**, divide to **4**: 8.2857. The 4th digit is 7, so it is **8.286**, not 8.285. Calculator allowed here only.

(a) $47 \div 9$, correct to 2 d.p. _____ (c) $26 \div 3$, correct to 3 d.p. _____

(b) $45 \div 7$, correct to 2 d.p. _____ (d) $100 \div 7$, correct to 3 d.p. _____

E4 Mr Bowen's homework

One of these four is right. Find it, and correct the other three.

(a) $6.4 \times 10^3 = 6.4000$ _____ (c) 2.996 to 2 d.p. = 2.99 _____

(b) $380 \div 10^2 = 38$ _____ (d) 0.83 to 1 d.p. = 0.8 _____

Section F · Where it all meets — reading a scale

In **A1** you were given a number and had to find the place. Here it is the other way round: you are given the place and have to find the number. A cylinder is a number line stood on its end, and a graph has two — and both of them ask the same question first: **how much is one small line worth?**

F1 How much is one small line worth?

Count the spaces, not the lines

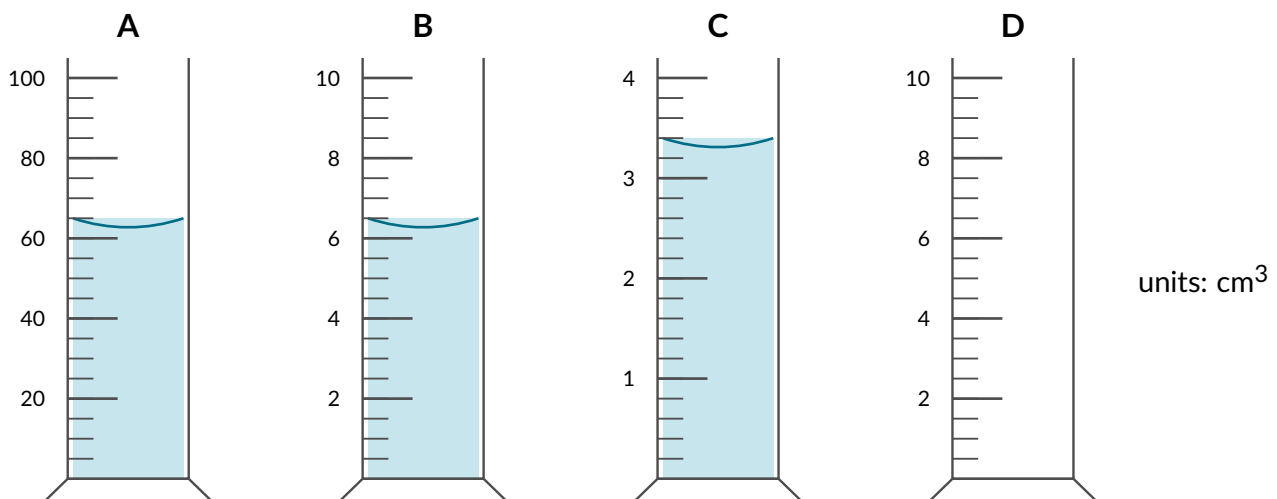
Look at **two numbered lines**. Count the **spaces** between them – the gaps, not the marks. Then **divide**. From **0 to 10** with **5 spaces**: $10 \div 5 = 2$, so one small line is **worth 2**.

A **scale** is the row of lines and numbers on an instrument. **Scales** is the machine you stand on to find your mass – a different word.

From	To	Spaces between them	One small line is worth
0	10	5	2
0	1	10	
20	30	4	
0	5	10	
50	100	5	

F2 Four measuring cylinders

Same small lines on all four; only the **numbers** differ. Read the bottom of the meniscus.

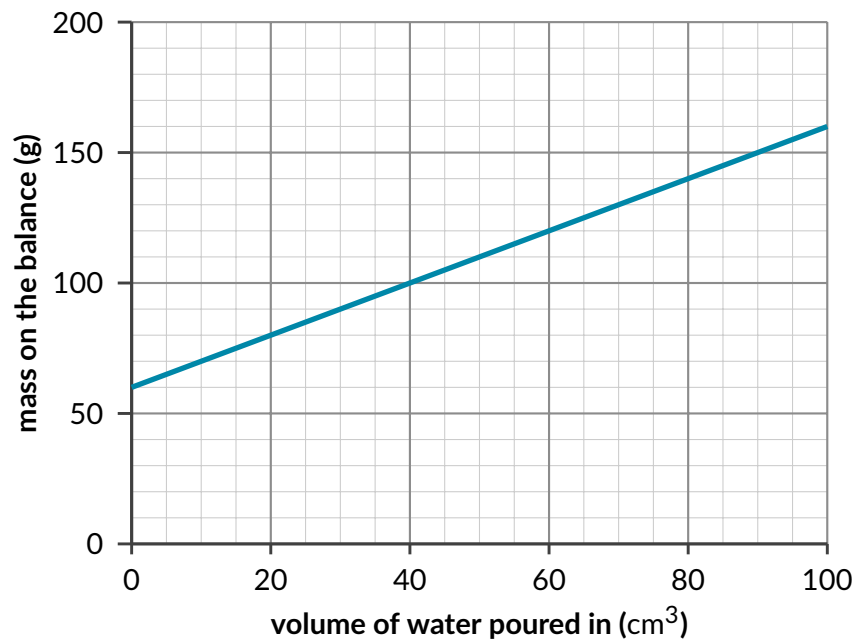


	One small line is worth	The volume of water is
A		
B		
C		
D		Draw the water at 8.5 cm³

(b) **A** and **B** are the same picture, and the water is at the same height. Explain why the two answers are not the same number.

F3 The same question, on a graph

Mr Bowen puts an **empty** beaker on a balance, then pours water in a little at a time and writes down the mass.



- (a) How much is one small line worth on the **across** axis? _____
- (b) How much is one small line worth on the **up** axis? _____
- (c) What is the mass when **40** cm³ has been poured in? _____
- (d) The balance reads **130** g. How much water is in the beaker? _____
- (e) What is the mass of the **empty** beaker? Say how you know.

- (f) Write a **formula** for the mass m grams when v cm³ of water has been poured in.

$m =$ _____

Answer Key – Teacher Copy

A1 Put each liquid on the scale

One small space is **0.2**, so every pH on the list lands exactly on a small line. Counting from the numbered line on its left:

A 2.4 – 2 lines past 2 · **B** 3.8 – 4 lines past 3 · **C** 5.2 – 1 line past 5 · **D** 7.0 – on the 7

E 8.2 – 1 line past 8 · **F** 9.6 – 3 lines past 9 · **G** 10.8 – 4 lines past 10 · **H** 12.6 – 3 lines past 12

Last column: A, B, C **acid** · D **neutral** · E, F, G, H **alkali**.

Mark the position, not the letter. A line within half a small space of the right place is right; a line on the wrong side of a numbered line is not. The error to look for is a student putting **B** at 3.8 by counting 8 small lines instead of 4 – they have read the 8 as eight spaces rather than eight tenths. Five spaces to a whole number means each one is two tenths, and that is the only arithmetic in the question.

G, 10.8 is the one most often put past 11. Counting 8 lines from 10 lands at 11.6. Same error as B, one column further left.

D, pure water at 7.0 is the row that separates the class, and the decimal is why it is written 7.0 rather than 7. It is not “a bit of both” and it is not “nearly acid”; it is **exactly** 7 and the word is **neutral**. Ask for that word specifically, and keep the answer to hand – C2(a) and C2(b) both turn on it.

E, sea water at 8.2 surprises them every year. The sea is slightly alkaline.

If a student writes *base* instead of *alkali* in the last column, correct it now: these liquids are all dissolved in water, so the exam word is **alkali**.

A2 Why the school buys both

Litmus has only two colours. pH 8 and pH 13 are *both* alkaline, so both turn red litmus blue, and there is no third colour for litmus to tell them apart with. He should use **universal indicator**, because it gives a different colour for every pH number, so it gives a number rather than just the word *alkali*.

This is the question that finds out who understood what an indicator *is*. The wrong answer to look for is “because 13 is stronger” – true, but it does not answer the question, which is about what litmus can *show*. Push for the words **two colours**.

A full-sentence answer is the requirement. “Litmus only has 2 colours” is the idea but not the sentence; the starters are printed for a reason.

A3 The end that nobody worries about

Wrong. A strong alkali is **corrosive** in exactly the way a strong acid is: it attacks skin, eyes and clothes. pH 13 is as far from neutral as pH 1 is. Danger is at **both ends** of the scale, not just the left-hand end.

This is the misconception of Science 2.8, and it was the class vote in the lesson – the majority chose pH 1. Expect the same split here.

The detail that makes it land is that the oven cleaner is **under his kitchen sink**, not in a lab. Students think of danger as a laboratory thing. Ask whose house has one.

Accept any answer that says both ends are dangerous *and* uses *corrosive* correctly. Do not accept “13 is a big number so it is strong” on its own – that is the student agreeing with the wrong sentence for a different reason.

A4 Neutralisation

(a) Lime powder, pH 12.

(b) The soil is an acid at pH 4. Lime powder is a base, so adding it neutralises the acid and the pH rises towards 7.

Pure water, pH 7 is the wrong answer worth discussing rather than marking. It is neutral, so students reason that adding a neutral thing makes the soil neutral. It does not — it dilutes it and the soil is still an acid. That is a genuinely good piece of thinking arriving at a wrong answer; say so before correcting it.

Vinegar and lemon juice are acids being added to an acid, which moves the pH the wrong way. A student who ticks one of those has not read *up to about 7*.

In (b) the phrase to insist on is **towards 7**. “It makes the pH bigger” is half the idea; neutralisation has a destination.

B1 Compound or mixture?

Mixtures: air · sand stirred into salt · mineral water · sea water.

Compounds: salt NaCl · water H₂O · carbon dioxide CO₂ · iron sulfide FeS.

Water and sea water on the same row is the whole question. H₂O is a compound; sea water is that compound with salt and other things stirred into it. Students who tick the same box for both have matched on the word *water*.

Air is the one they most often call a compound, because it is invisible and feels like one substance. Ask what you can take out of it — oxygen, for a hospital cylinder.

Mineral water is the easy one once air has been settled, and it is on the packet as the reward for getting air right.

B2 Iron and sulfur, twice

Stirred, the iron and the sulfur are a **mixture**: the atoms have not joined, so the iron is still iron and a magnet pulls it out. Heating made the atoms **bond** together, and what is left is a **new substance**, iron sulfide. It is a **compound**, and it is not magnetic, because it is not iron any more.

Mark for the word **new substance**. The near-miss answer is “the heat melted them together”, which is a mixture description wearing a compound’s clothes — the student thinks the iron is still in there, just stuck. It is not.

The magnet is the evidence and the answer should say so. A student who explains bonding but never mentions the magnet has answered a different question.

Three writing lines, because this needs three sentences: what it was, what heating did, what it is now.

C1 Say it in English

$9 > 2$ — “9 is greater than 2”. $-4 < 1$ — “-4 is less than 1”. $-7 < -3$ — “-7 is less than -3”.

$-7 < -3$ is the row that goes wrong, and the error is Unit 1’s, not this unit’s: 7 is bigger than 3, so the student writes $>$. Send them to the number line, or to the thermometer — -7 degrees is colder.

The English column is the point of the question. Accept only a full sentence with **is**: “-4 less than 1” is not it. They have known these signs since Grade 1; what they have never done is say them out loud in English, left to right.

C2 Which liquids does the inequality mean?

(a) $p < 7$: A, B, C (b) $p > 7$: E, F, G, H

(c) $p > 9.6$: G, H (d) $p < 3.8$: A

(e) $p > 5$ and $p < 9$: C, D, E (f) $p > 12.6$: none

Four of the six turn on a liquid whose pH is exactly the number in the inequality, and that is why this question exists. In (c) baking soda is 9.6 and $9.6 > 9.6$ is false, so F is out. In (d) orange juice is 3.8, so

B is out. In (f) bleach is 12.6, so nothing on the list works *at all* — and the instruction line warns that one of the six has no answer. A student who keeps F in (c) will keep H in (f); mark those two together and hand them back at once.

In (a) and (b), notice that **D, pure water, appears in neither**. 7.0 is not less than 7 and not greater than 7. If a student has put D in both, they have read $<$ and $>$ as “on the acid side” and “on the alkali side”.

(e) is the two-inequality question and the only one where a liquid must satisfy both. A student who writes C, D, E, F has checked $p > 5$ and stopped reading — 9.6 is not less than 9.

This is also a science check: (a) is exactly the list of acids and (b) is exactly the list of alkalis. If their A1 table disagrees with their C2 answer, one of the two is wrong and they can find out which without you.

C3 Now write the inequality

(a) $p > 7.2$ (b) $r < 5.6$ (c) $s > 5$ (d) $c < 2$ and $c > 12$

The vocabulary is the question: **above** and **more than** are both $>$; **below** and **under** are both $<$. Four different English words, two signs.

(d) needs *two* inequalities, and it says so. A student who writes $2 > c > 12$ has invented something that cannot be true; do not mark it wrong so much as ask them to read it aloud — it says 2 is greater than 12.

The decimals in (a) and (b) are not there to be hard. They are there so that the class sees an inequality does not need a whole number in it, which is what makes C5(d) answerable.

C4 Read the line, then draw the line

(a) (i) $x < 5$ (ii) $x > -3$

(b) (i) open circle at 4, arrow **left** (ii) open circle at -2 , arrow **right**

In (a) the arrow direction is the answer and the circle is where it starts. Students who read the arrow-head's number instead of the circle's write $x < -1$ for (i).

In (b), check the **circle is open**. A filled circle is a different inequality ($\leq$), which is next year, and a student drawing it here has copied a dot from somewhere.

(a)(ii) crosses zero, which is the only reason it is on the packet.

C5 The smallest one that works

(a) **7** (b) -5 (c) **0** (d) **3**

(b) is the negative-direction trap: *largest* feels as though it should mean *to the right*, so students write -3 . Less than means **further left**, and the largest integer below -4 is the one immediately to its left.

(c) answers **0**, and students who think zero is not really a number leave it blank.

(d) is the decimal one: the circle sits between 2 and 3, so the smallest *integer* that works is 3, not 2.5 and not 2.

D1 Put it back in and see

$x - 7 = 5$, $x = 2$: **wrong**, $x = 12 \cdot 3y = 24$, $y = 8$: **right**

$m + 9 = 15$, $m = 24$: **wrong**, $m = 6 \cdot k - 6 = 10$, $k = 4$: **wrong**, $k = 16$

Every wrong answer here is the *same* error: the student did the operation they could see instead of the inverse. 2 is $7 - 5$; 24 is $15 + 9$; 4 is $10 - 6$. Say that out loud — three mistakes, one habit.

The fix is the one line of checking the box shows. A class that substitutes can settle its own arguments, and every workbook question in 2.5 asks them to.

D2 Two steps

- (a) $26 - 5 = 21$, $21 \div 3 = 7$ (b) $25 + 7 = 32$, $32 \div 4 = 8$ (c) $47 - 12 = 35$, $35 \div 5 = 7$
 (d) $15 + 9 = 24$, $24 \div 2 = 12$ (e) $40 - 4 = 36$, $36 \div 6 = 6$ (f) $45 + 3 = 48$, $48 \div 8 = 6$

Mark the **middle line**. A student who divides first gets a number that looks like an answer, and nothing in the final figure shows what went wrong.

The dividing-first error on (a) gives $26 \div 3$, which is not a whole number — and that is useful. Tell them: if the first step does not divide exactly, you almost certainly did it in the wrong order.

D3 You are given the formula. Find the letter.

- (a) $4s = 52$, $s = 13$ (b) $8n + 15 = 95$, $80 \div 8$, $n = 10$ (c) $60t = 480$, $t = 8$
 (d) $30h = 420$, $h = 14$ (e) $60 + v = 145$, $v = 85$ (f) $2n - 6 = 38$, $44 \div 2$, $n = 22$

The instruction is to write the equation first, and that is what to mark. A student who writes only “13” has probably divided 52 by 4 for good reasons, but the packet is teaching a formula becoming an equation, and the equation is the evidence.

(a), (c) and (d) are one step; (b) and (f) are two. The mixture is deliberate — the formula does not announce which kind it is, and deciding that is the skill.

(e) is the formula the graph in F3 is drawn from. If a student notices, that is the best thing that will happen on this packet; F3(f) asks for it from the other end.

D4 Word problems

- (a) $6w + 40 = 340$; $340 - 40 = 300$, $300 \div 6 = 50$. **50 cm³** in each beaker.
 (b) $(f + 4) \div 2 = 6$; $6 \times 2 = 12$, $12 - 4 = 8$. He had **8 fish**.

(a) is the one where the English decides everything. *40 is left in the jug* means 40 was never poured, so it is +40 on the beaker side, and the students who write $6w = 340$ have skipped the clause. Make them point at the words.

(b) is the only question on the packet where the last operation is a division, so *undo the last step first* means multiply. Students reverse it and get $(6 - 4) \times 2 = 4$. The check is the cure: 8 fish, buy 4, that is 12, the cat eats half, 6 left.

Read (b) completely straight. Do not explain the cat.

E1 Move the digits

- (a) 370 (b) 4500 (c) 260 000 (d) 9.4 (e) 0.063 (f) 0.0005

(e) and (f) are where the zeros go missing. $63 \div 10^3$ is *three* places right: 6.3, 0.63, 0.063. A student who writes 0.63 has moved two.

(f) is the extreme case — four places from a single digit, so three zeros have to be written before the 5 arrives.

(b) is the one that catches the folk rule: “add three zeros” gives 4.5000, which is just 4.5.

E2 Round it

Top row, to 1 d.p.: 7.3 · 12.9 · 1.0 · 30.0

Bottom row, to 2 d.p.: 5.38 · 0.10 · 8.24 · 4.00

Four of the eight are the same question, and it is the one from the lesson: the digit carries and the trailing zero must stay. 0.952 is 1.0 not 1; 29.96 is 30.0 not 30; 0.0961 is 0.10 not 0.1; 3.997 is 4.00 not 4. The zero is what shows how many places you rounded to, and dropping it loses the mark even though the number is the same size.

8.2449 is the opposite trap and the only one on the packet: the third decimal is 4, so it rounds *down* to 8.24. Students who look at the *fourth* digit, see 9, round the 4 up to 5 and then round again get 8.25. Rounding happens once, from the original number.

E3 One place further

(a) $5.2222 \dots = 5.22$ (b) $6.4285 \dots = 6.43$ (c) $8.6666 \dots = 8.667$ (d) $14.2857 \dots = 14.286$

(b) and (d) are the ones that need the extra digit. (b) to two places means dividing to three: 6.428, and the 8 rounds the 2 up to 3. A student who stops at two places sees 6.42 and never finds out.

(a) and (c) are the recurring ones, and (c) is the reminder that 6 always rounds up — 8.6666 to 3 d.p. is 8.667.

E4 Mr Bowen's homework

(a) **wrong** — 6400. The digits move; you do not write zeros on the end of a decimal.

(b) **wrong** — 3.8. Two places right from 380 is 38, then 3.8.

(c) **wrong** — 3.00. The 6 rounds the second 9 up, which carries all the way, and all three digits of 3.00 are needed.

(d) **right**.

(c) is the hardest item in Section E: it is the carry *and* the trailing zeros in one number, and the answer no longer looks like the question.

The instruction says one of the four is right. A student who corrects all four has not read it; a student who corrects none has decided Mr Bowen is reliable.

F1 How much is one small line worth?

0 to 1, 10 spaces: **0.1** 20 to 30, 4 spaces: **2.5** 0 to 5, 10 spaces: **0.5** 50 to 100, 5 spaces: **10**

The 20 to 30 row is the one to watch: the gap is 10, not 30. Students subtract nothing and divide 30 by 4. Say **the gap** rather than **the numbers** when you hand this back.

The 0 to 5 row is the one that proves the answer depends on *both* numbers: the gap is bigger than the first row's and the answer is smaller, because there are twice as many spaces.

Counting lines instead of spaces gives 6 spaces where there are 5, and it is the error this whole section exists to kill. A scale from 0 to 10 in fives has six marks and *five* gaps.

F2 Four measuring cylinders

A: one line = 5 cm^3 , water = **65** cm^3

B: one line = 0.5 cm^3 , water = **6.5** cm^3

C: one line = 0.2 cm^3 , water = **3.4** cm^3

D: one line = 0.5 cm^3 ; 8.5 cm^3 is **17 small lines up**, which is one line above the 8.

(b) The small lines are worth different amounts. On A each line is 5 cm^3 and on B each line is 0.5 cm^3 , so the same height is ten times as much on A as on B.

(b) **is the question, and A and B are drawn identically on purpose.** The two answers differ by exactly a factor of ten, which is Section E standing in a cylinder. If a student has written 65 for both, they read the picture and never read the numbers — which is the single commonest error in practical work.

C is the hardest read: five spaces per number, so one line is 0.2, and 17 lines is 3.4. Students who assume 0.5 because B was 0.5 get 8.5. Make them count the spaces between two numbers on *that* cylinder.

In D, check the level is drawn one line *above* the 8, not on it, and that the meniscus dips.

F3 The same question, on a graph

(a) 5 cm^3 (b) 10 g (c) 100 g (d) 70 cm^3

(e) 60 g — it is where the line meets the up axis, which is the mass when no water has been poured in at all.

(f) $m = 60 + v$

(a) and (b) are F1 on a graph, and they must be done before (c): four spaces per 20 across, five spaces per 50 up. A student who guesses that every small line is 1 gets everything after this wrong, so mark (a) and (b) first.

(e) is the best question on the packet. The graph never says “empty beaker” and there is no point marked “0 water” to read — the line simply starts at 60, and the reason it starts there is the beaker’s own mass. Students look for a word. Accept any answer that finds 60 *and* explains it as the mass before any water went in.

(f) is D3(e) from the other end. There, they were handed $m = 60 + v$ and asked for v ; here they are shown the line and asked for the formula. Hand the two back together — the graph and the formula are the same object, and this packet has no better moment in it.